

Cache Memory Mapping - Practice Questions#1

Question 1:

An embedded processor uses 32-bit words for accessing its 16MB main memory system. A program that performs a matrix scan operation resides completely in set# 0 of a 4KB direct mapped cache memory that has a block size of 16 words. The program does the operation $C[i] = \text{Max}\{A[i], B[i]\}$ with the help of a loop that iterates the value of i from 0 to 63. The largest among $A[i]$ and $B[i]$ for each i gets copied to $C[i]$. A , B , and C are one dimensional arrays of 64 words each and are assigned with physical memory addresses starting from 0x412888, 0x72A900, and 0x236770, respectively. Data words of A , B and C are brought to the cache upon demand and are referred once per iteration in the order B , A and C . Assume that the cache is initially empty except the set# 0 that holds the program.

1. Identify the tag, set index and offset split-up of the physical address.
2. Identify the number of data misses encountered during the execution of the program.
3. Which all words of A , B and C are residing in the cache after the completion of the matrix operation?

Question 2:

A program stored in a 16MB main memory system consists of a main routine M which in turn calls two subroutines; P and Q . A processor with a word length as well as instruction length of 16 bits fetches and executes the above program and its associated data. M has 12 instructions ($M1-M12$) stored continuously from the main memory address 0x78A4FC, P has 4 instructions ($P1-P4$) stored continuously from the main memory address 0x222454 and Q has 6 instructions ($Q1-Q6$) stored continuously from the main memory address 0xFCE4F8. $M5$ to $M12$ (both inclusive) is a loop that iterates 20 times. $M6$ is a subroutine call to P and $M10$ is a subroutine call to Q . The last instruction of both P and Q are subroutine returns back to M . The M , P and Q together access three 4-word integer arrays ($F1$, $F2$ and $F3$) that are assigned physical addresses $F1 \rightarrow 0x400500$, $F2 \rightarrow 0x411500$ and $F3 \rightarrow 0x444600$, respectively. The processor has a 16KB direct mapped unified cache with a block size of 256B. Assume the cache is initially empty. Ignore OS level interruption and subsequent cache impact on context switching.

1. Give the split of tag, set index and offset fields of physical address.
2. How many compulsory misses and conflict misses occurred during the execution of the program?
3. List out the cache block numbers (in decimal) that are non-empty after the execution of the program. Which all instructions of M , P and Q are residing in the cache at the end of execution of the program?
4. What is the MPKI value (misses per kilo instruction) of the program?